

Network fundamentals

What is a network

What is a network

- A computer network is a group of at least two devices that can communicate (share data) with each other over the network.
- The devices can be connected through both physical and/or wireless connections.
- The scale can range from just 2 computers sharing files to the complete internet, interconnecting billions of devices.



Network communication

- An application on one device wants to exchange or request information (files, raw data, data stream) with one or more other devices.
- The devices operating system will put the information on the network
- The network will route the data to the correct destination
- A lot of underlying mechanisms (protocols) will enable the network and the devices to communicate

Communicatoin between applications

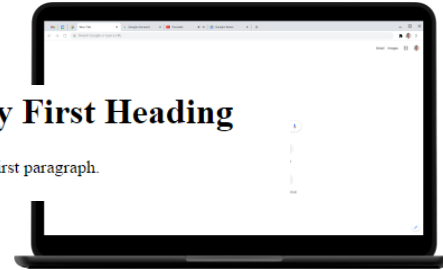


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<p>My first paragraph.</p>

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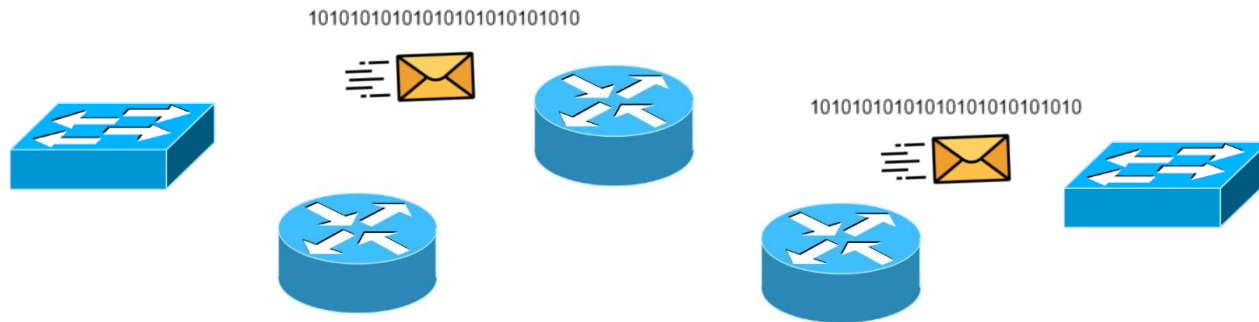
NGINX



Linux

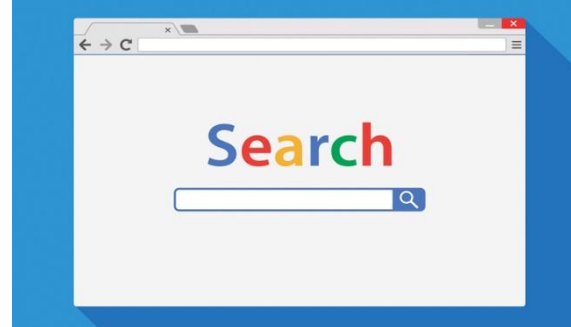


Windows



How networks impact our daily lives

- World wide web
- Home entertainment
 - Video streaming
 - Video sharing
 - Gaming
- Instant messaging
 - Text/voice/video
- Content sharing
 - Blog/Vlog
 - Podcasts
 - Wiki
- Smart home/car/city
- Internet of Things
- Information available everywhere
- Smartphone, laptop



How networks impact our Learning

- Virtual Learning Spaces
- Classrooms
- E-learning
- On-Demand Video
- Collaborative



How networks impact our Businesses

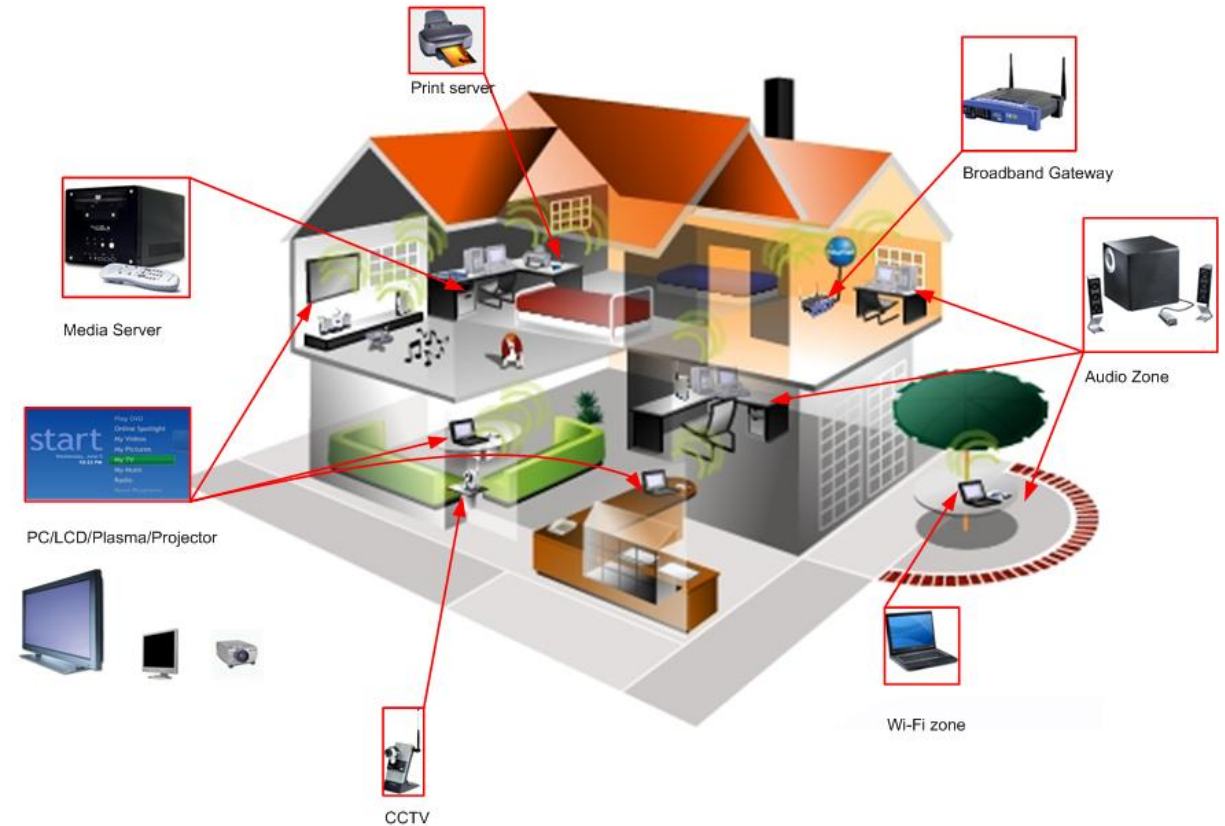
- Intranet
- VoIP
- Video conferencing
- New business models
- IoT
- Smart factory



Networks of different sizes

Small home networks

- Internet connection
- Wireless access point
- Few devices



Networks of different sizes

Medium sized office network

- (Multiple) internet connections
- Wireless network
- Larger number of devices
 - Computers
 - Laptops
 - Printers
 - Servers
 - VoIP
- Security and management



Networks of different sizes

Global enterprise networks

- Many locations all over the world
- Hundreds or thousands interconnected



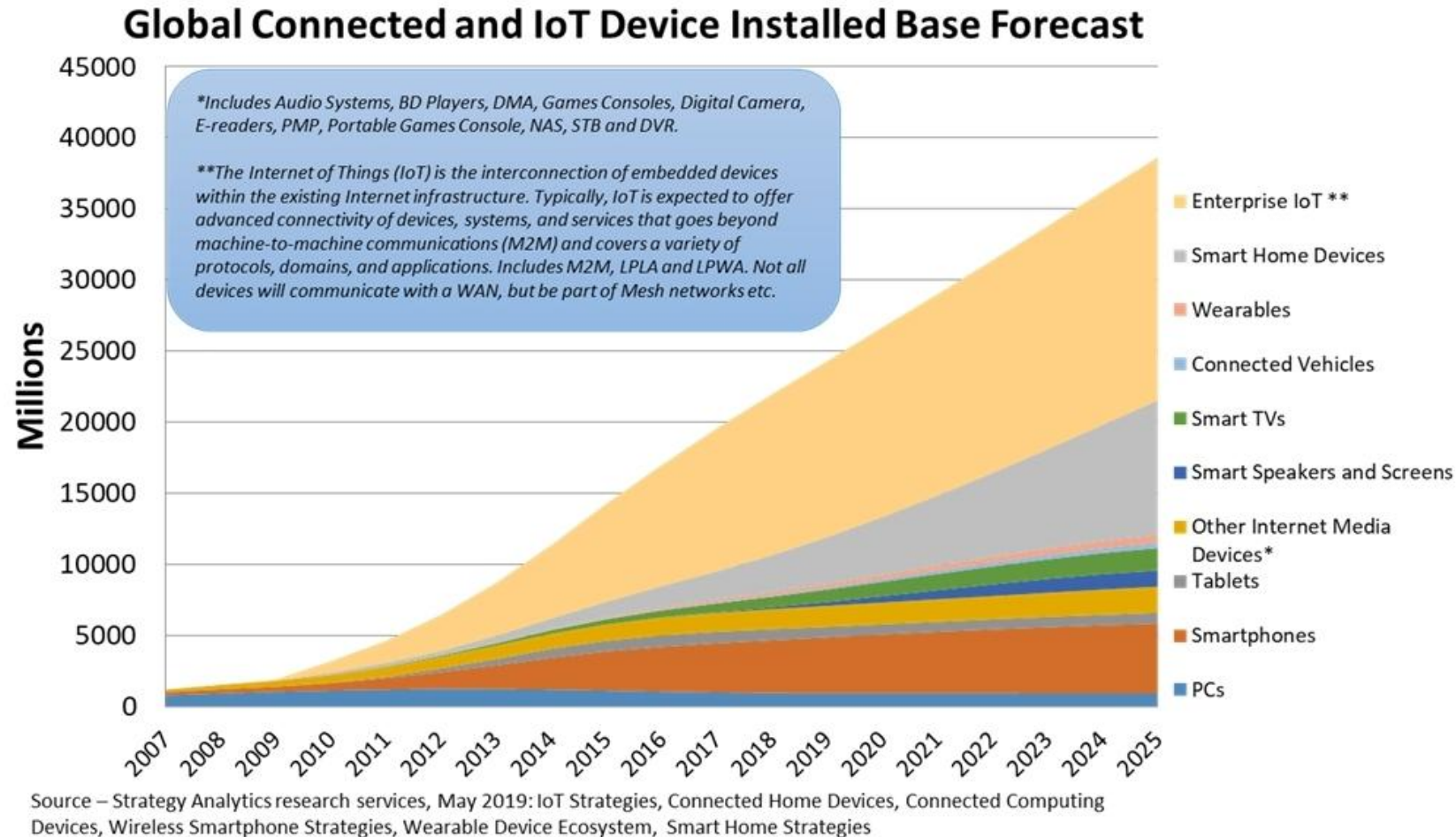
Networks of different sizes

Internet

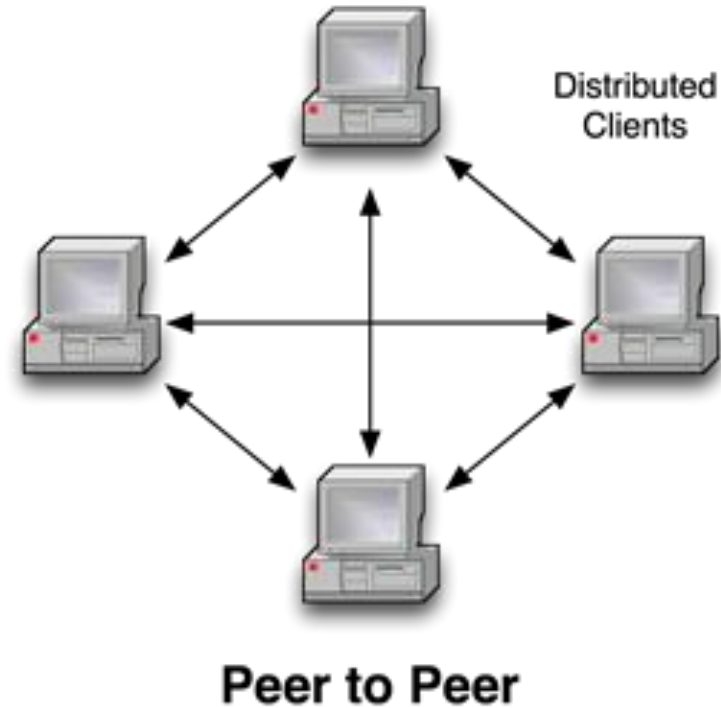
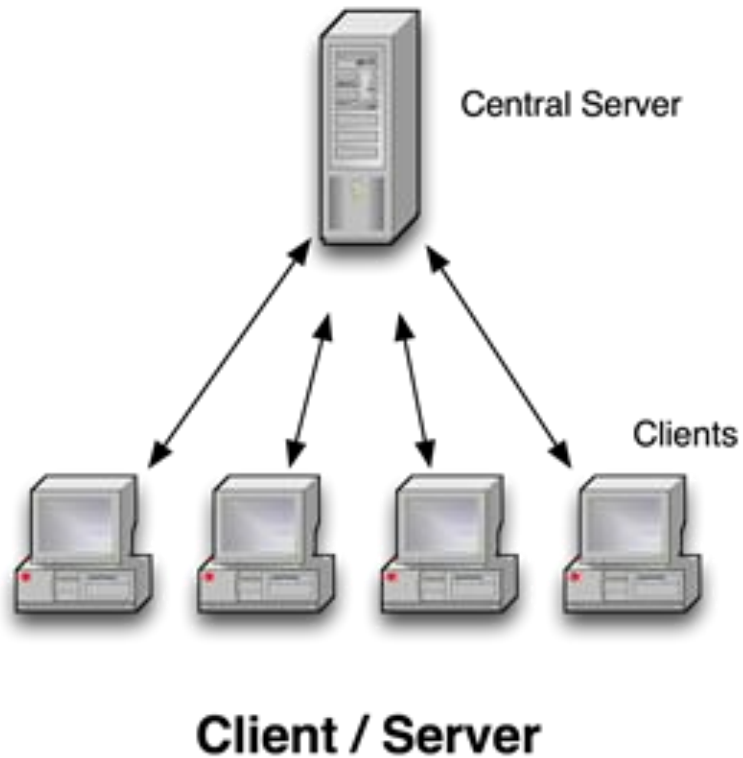
- Global interconnected network
- Interconnection of home/small business



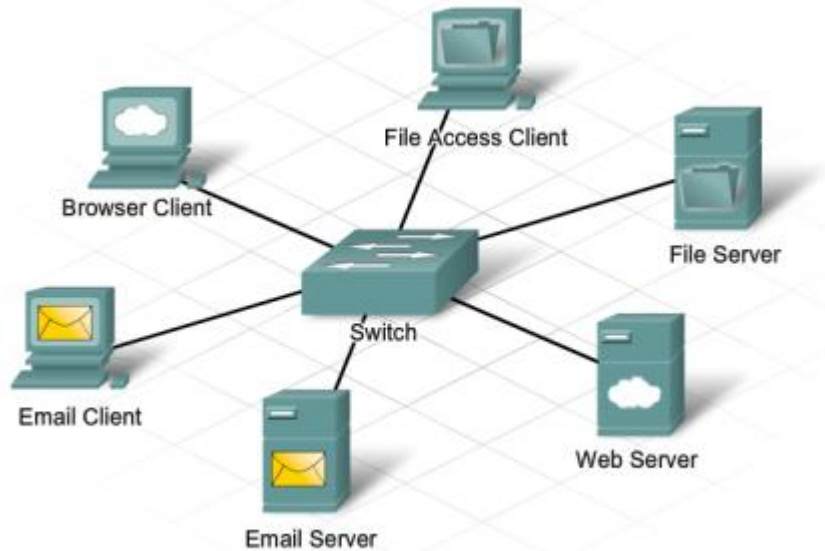
Number of connected devices



Ways to organize communicate



Clients and Servers

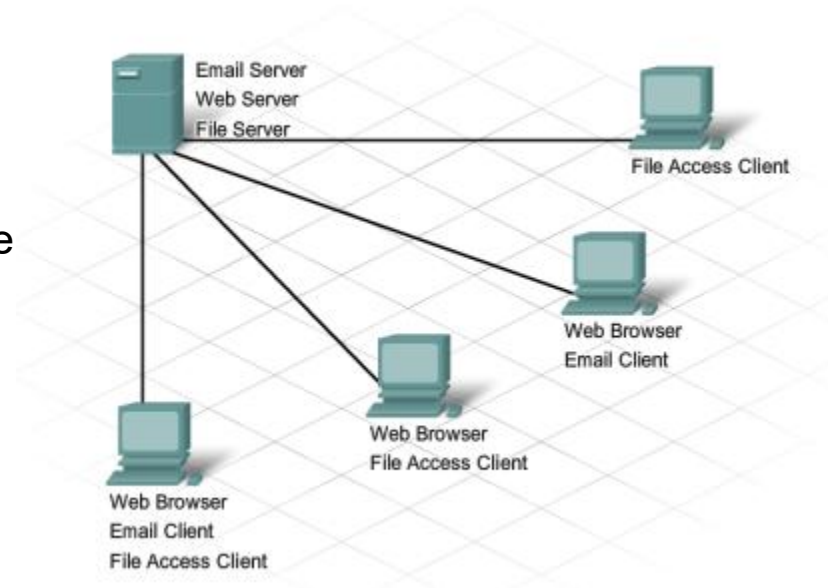


A computer with server software can provide services simultaneously to one or many clients. A single server can run multiple types of server software.

An example of server software is a webserver like Apache, Nginx, IIS, ...

Clients are computer hosts that have software installed that enable them to request and display the information obtained from the server. A single computer can run multiple types of client software.

An example of client software is a web browser, like Brave, Firefox, Chrome, Safari, ..



A single device can run multiple types of client and server software simultaneously.

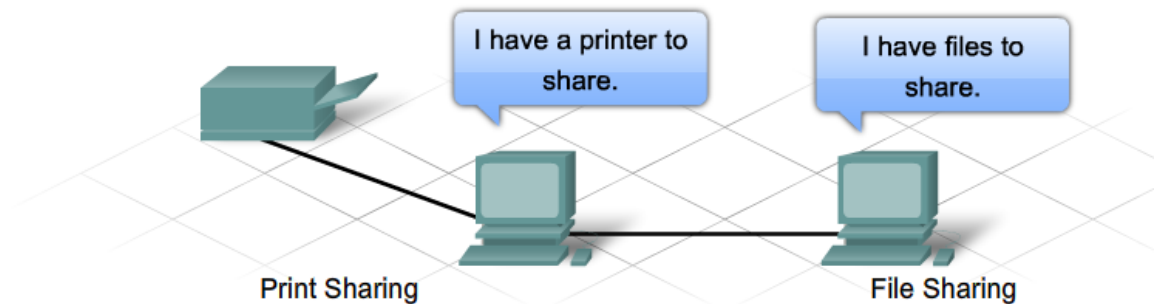
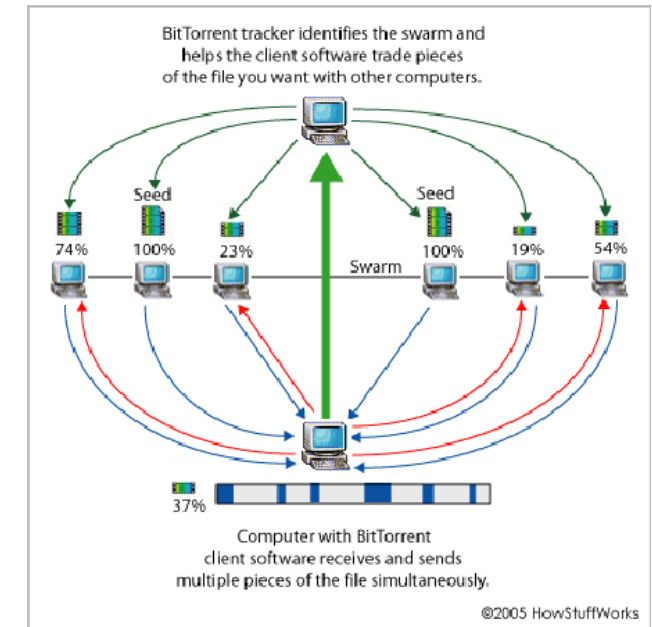
Such a device will be a client and a server on the network, depending on the specific service.

For example, you can use a web browser on a server that has a web server installed.

Peer-to-Peer network

All devices can act both as client and server

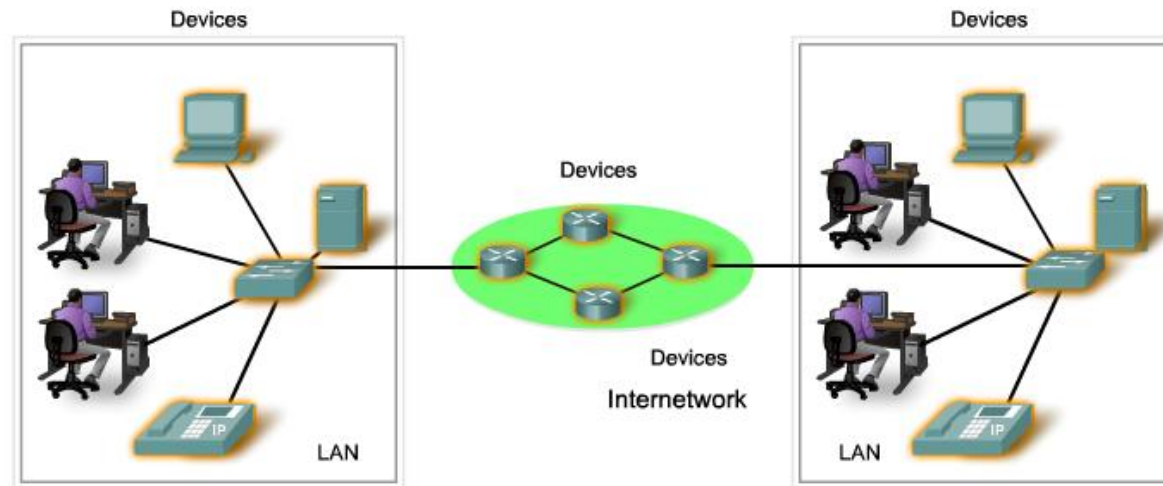
- Easy to set up
- Low complexity for small networks
- Low cost
- No centralized administration
- Not scalable
- Difficult to keep secure



Components of a Network

There are three categories of network components:

- Devices
- Media
- Services (& processes)



End Devices

Some examples of end devices are:

- Computers (workstations, laptops, file servers, web servers)
- Network printers
- VoIP phones
- Security cameras
- Mobile devices (smartphones, tablets, smartwatches)
- Internet of Things (sensors, lights, fridges)



Network Infrastructure Devices

Examples of intermediary network devices are:

- Network Access Devices (switches, and wireless access points)
- Internetworking Devices (routers)
- Security Devices (firewalls)

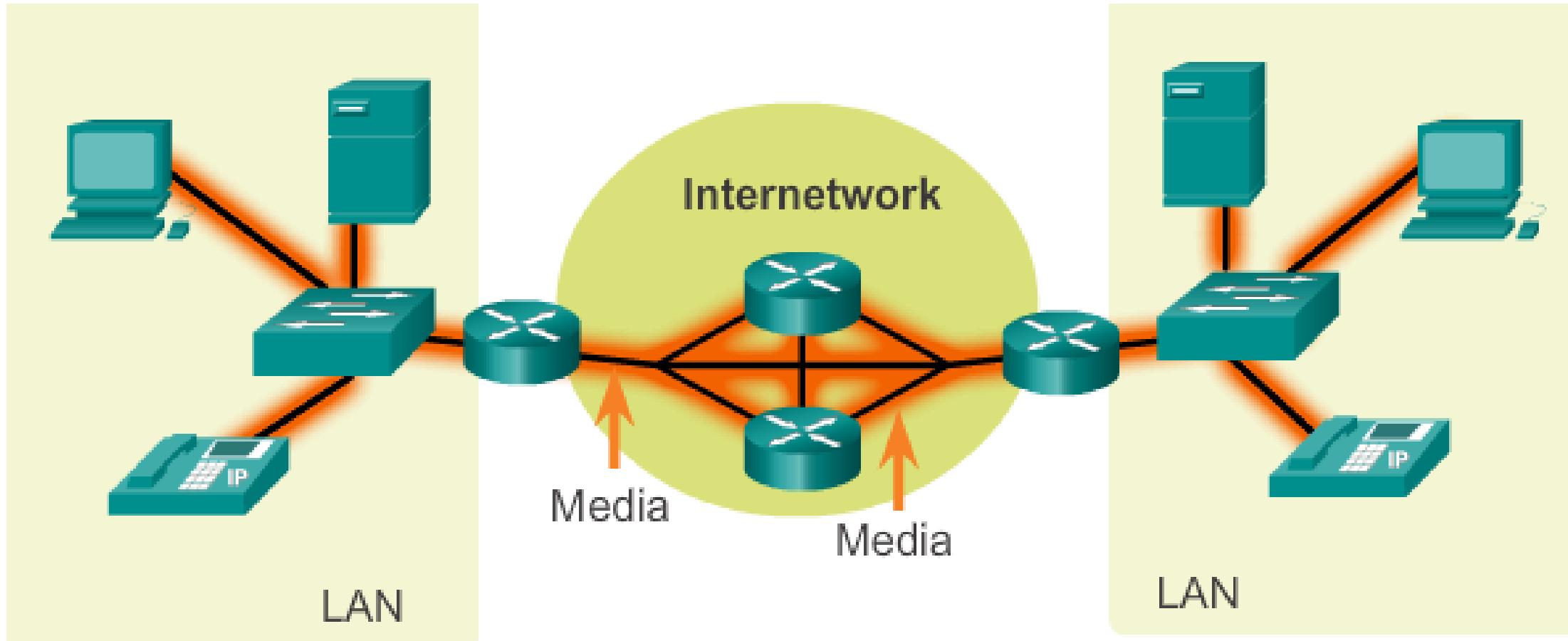


Network Infrastructure Devices

Role of an intermediary device

- Processes running on the intermediary network devices perform different functions:
 - Regenerate and retransmit data signals
 - Maintain information about what pathways exist through the network and internetwork
 - Notify other devices of errors and communication failures
 - Direct data along alternate pathways when there is a link failure (redundancy)
 - Classify and direct messages according to QoS priorities
 - Permit or not permit the flow of data, based on security settings (access control lists ACL)

Network Media



Network Media

The criteria for choosing network media are:

- The distance the media can successfully carry a signal
- The environment in which the media is to be installed
- The amount of data and the speed at which it must be transmitted
- The cost of the media and installation

Copper



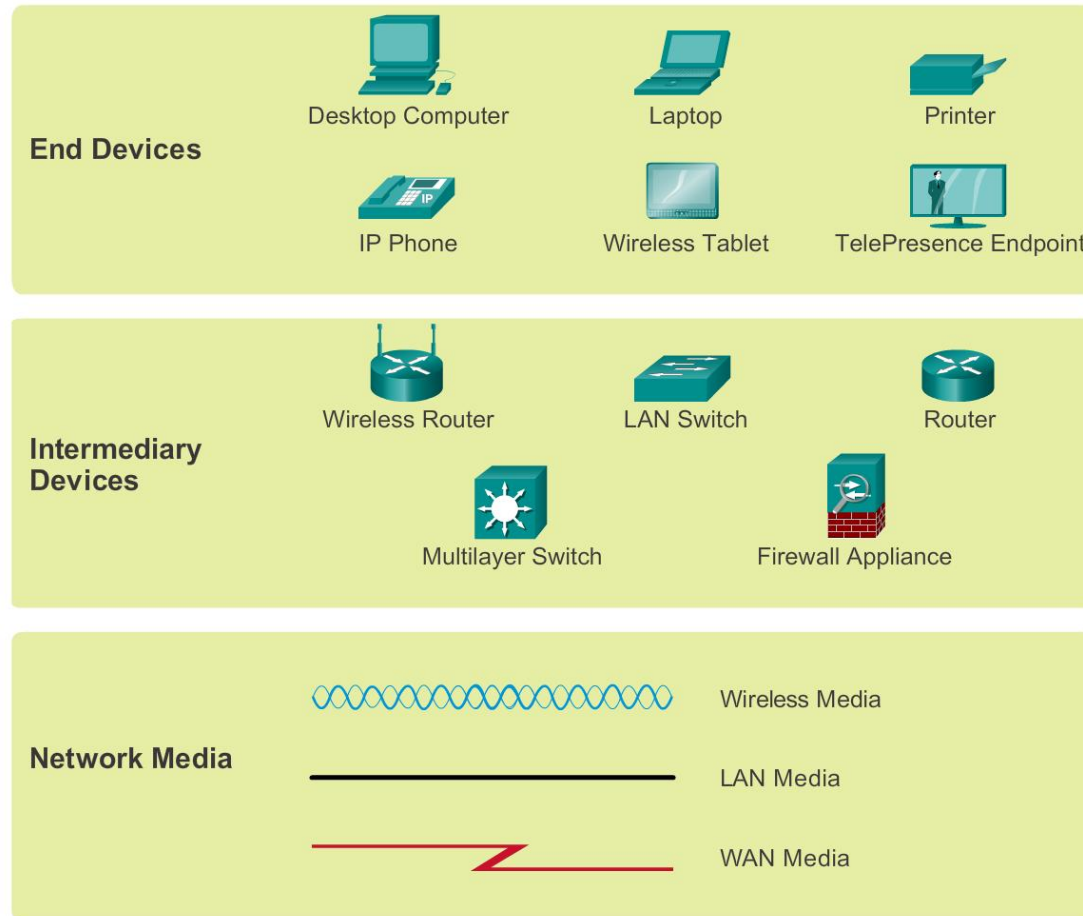
Fiber Optic



Wireless



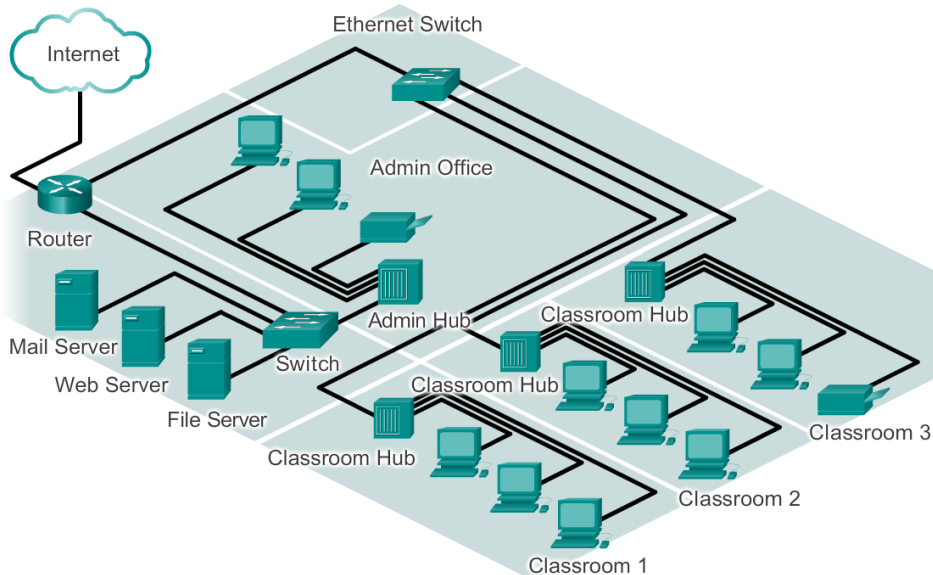
Network Representations



Topology Diagrams

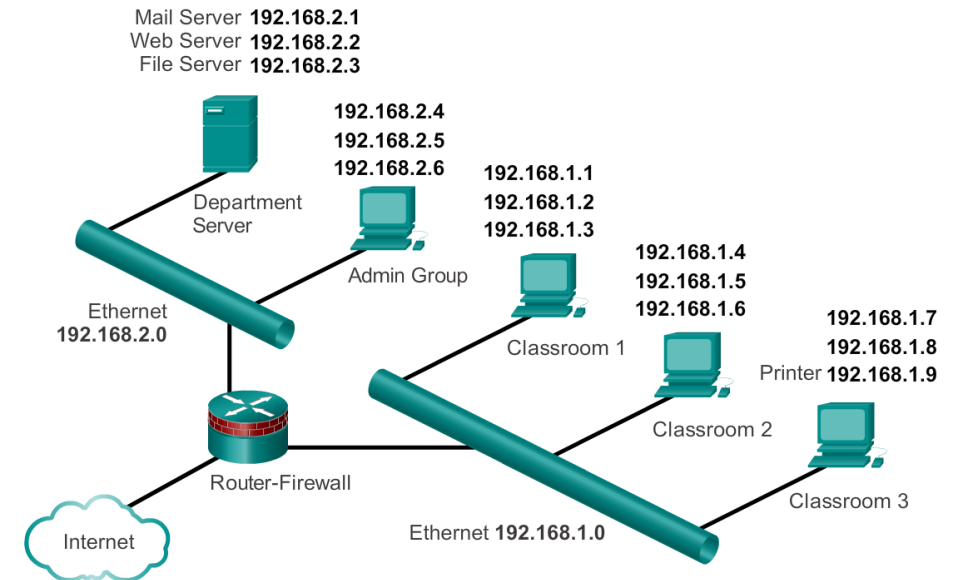
- **Physical topology** diagrams - Identify the physical location of intermediary devices, configured ports, and cable installation.

Physical Topology



- **Logical topology diagrams** - Identify devices, ports, and IP addressing scheme.

Logical Topology



Types of Networks

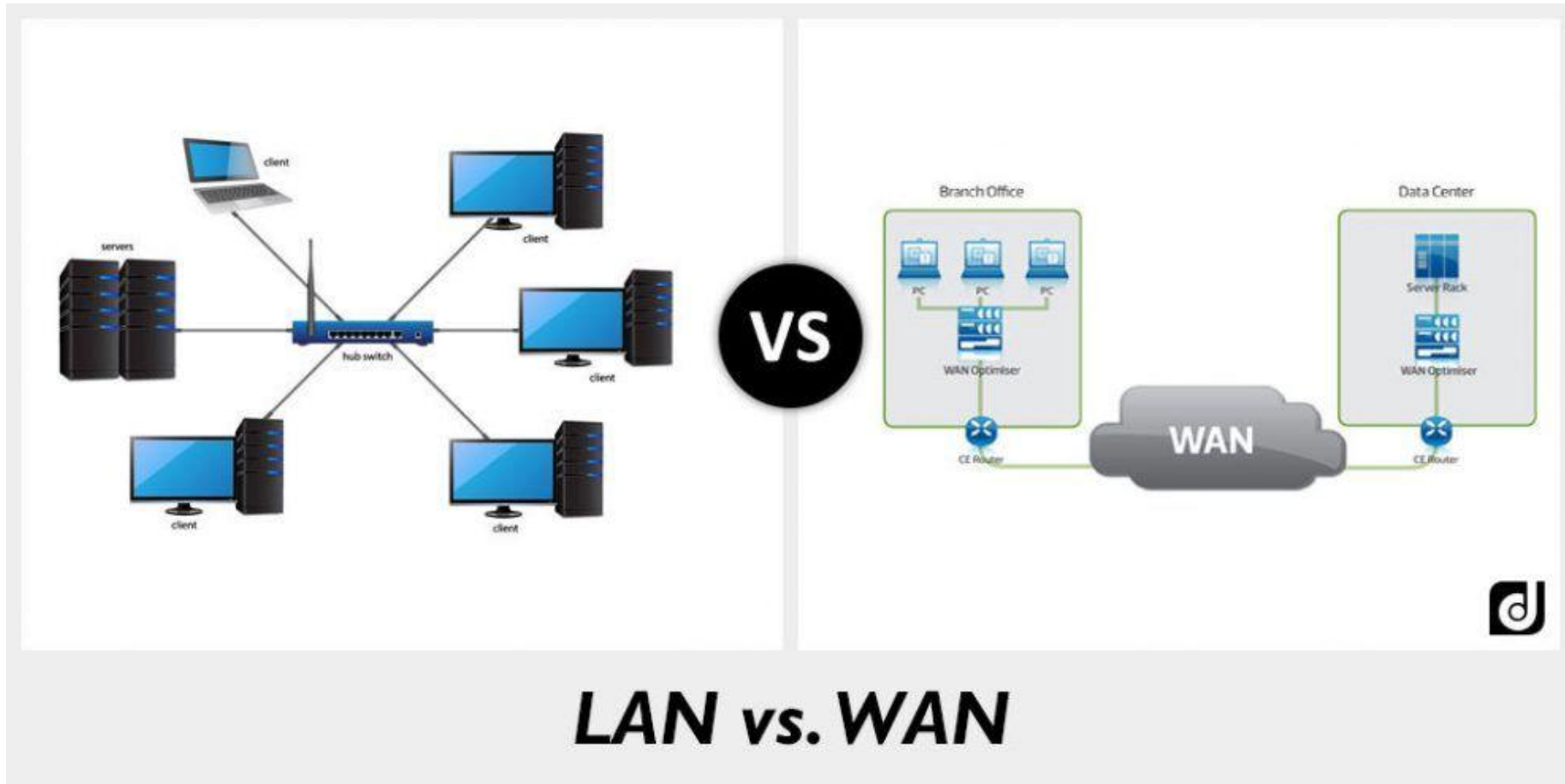
The two most common types of network infrastructures are:

- **Local Area Network (LAN)**
- **Wide Area Network (WAN).**

Other types of networks include:

- Metropolitan Area Network (MAN)
- Wireless LAN (WLAN)
- Storage Area Network (SAN)

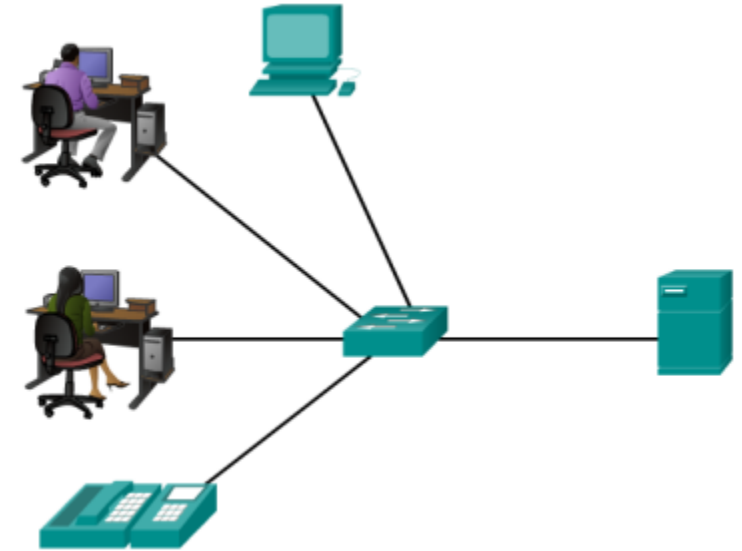
LAN - WAN



Local Area Networks (LAN)

Local Area Network (LAN)

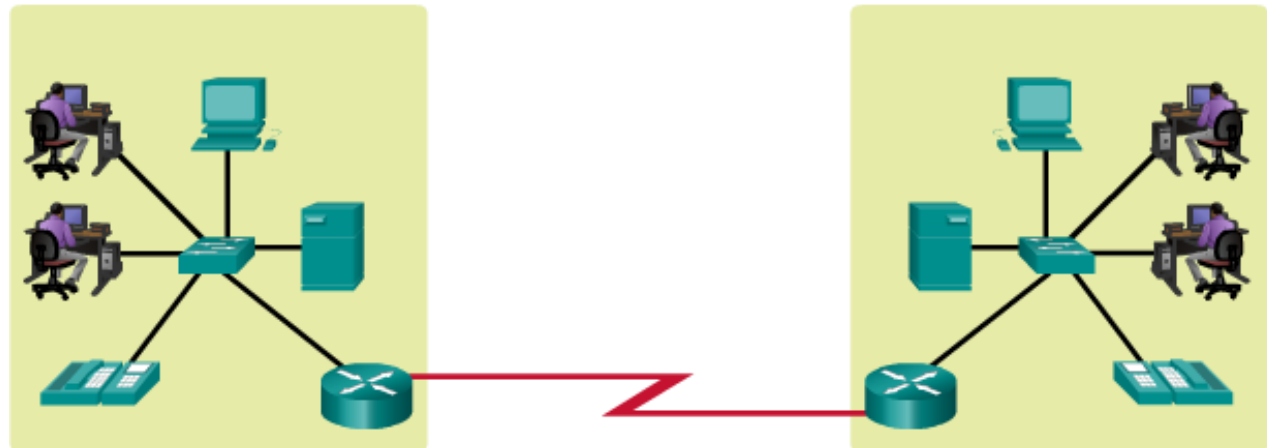
- A network infrastructure that provides access to users and end devices in a small geographical area.
- A LAN is usually administered by a single organization or individual.
- LANs provide high speed bandwidth
- Protected from the internet with firewall



Wide Area Networks (WAN)

Wide Area Network (WAN)

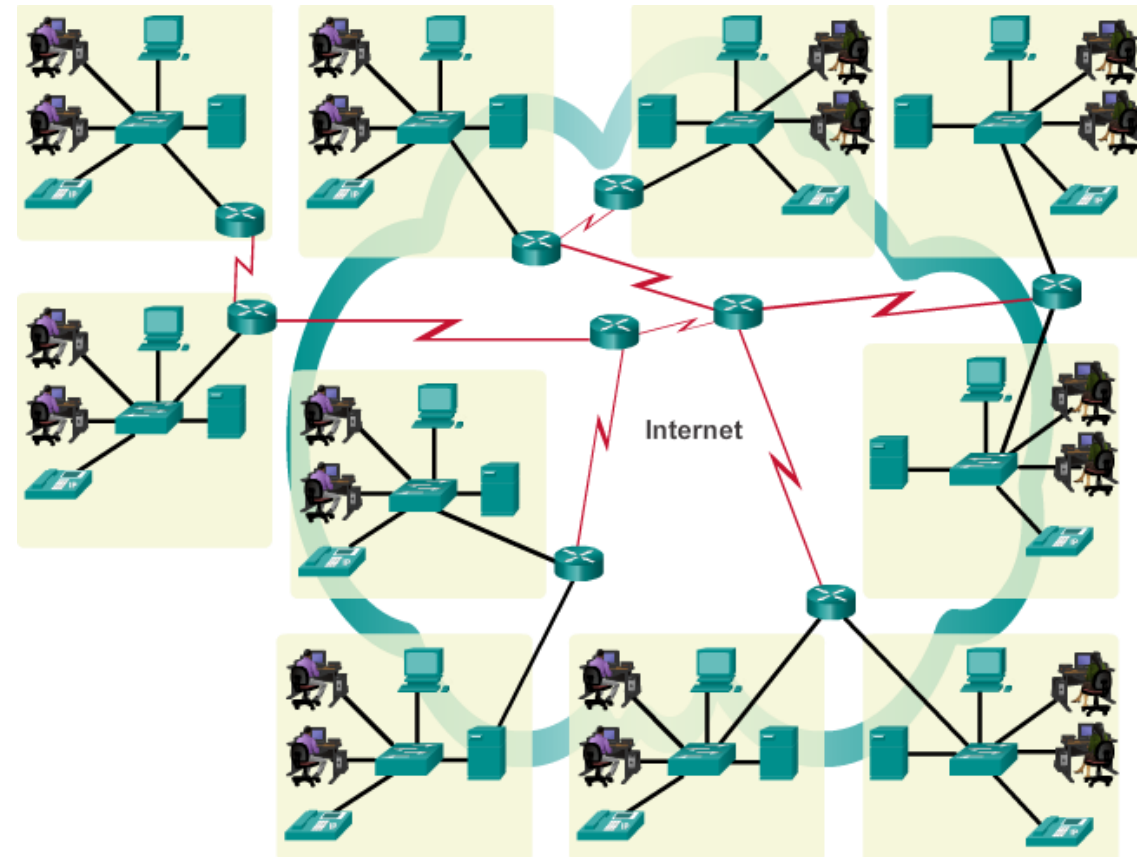
- A network infrastructure that provides access to other networks over a wide geographical area.
- WANs are usually administered by multiple service providers.
- WANs typically provide slower speed links between LANs.



The Internet

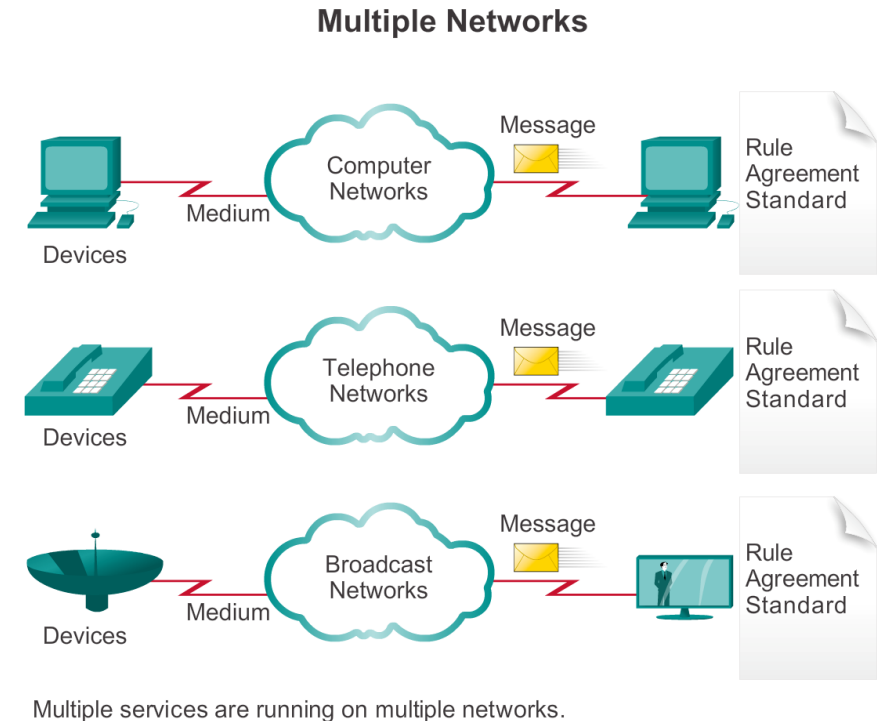
Internetwork

- An internetwork is defined as a global mesh of interconnected networks (WANs + LANs), example: the internet



Multiple networks

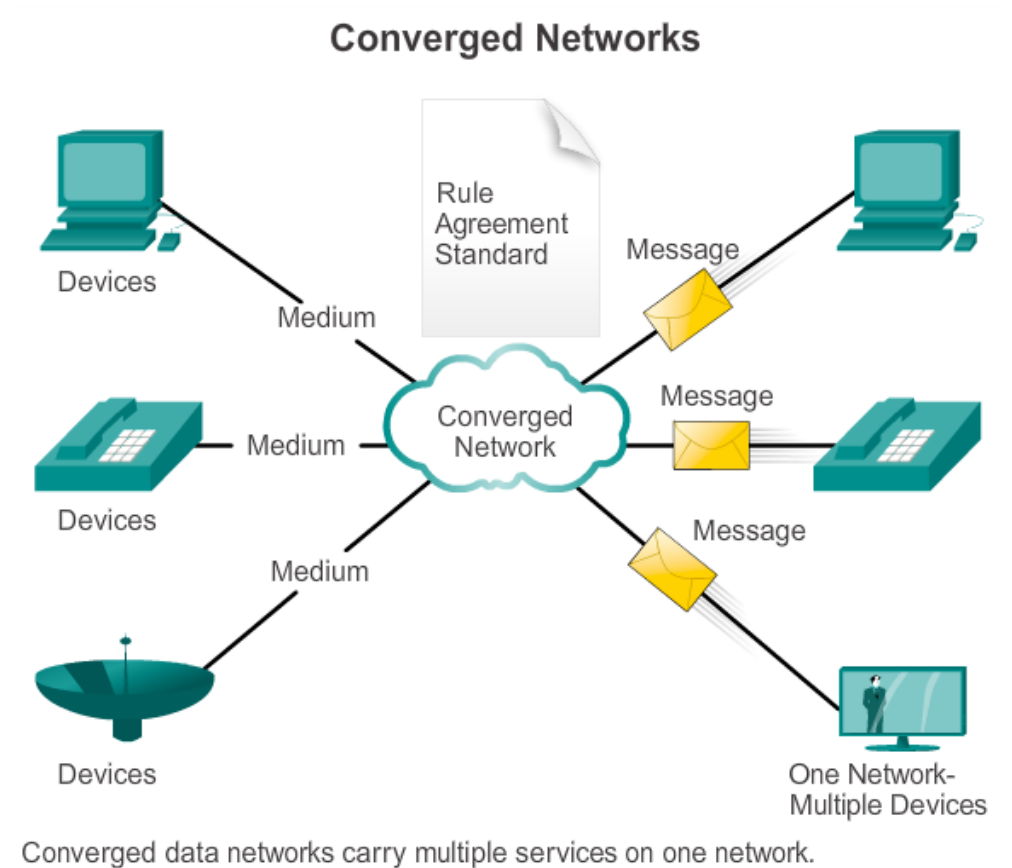
In the past classrooms were cabled for the data network, telephone network, and video network for televisions. These **separate** networks were separated; meaning that they **could not communicate** with each other



Converged networks

Different developments: Converged networks

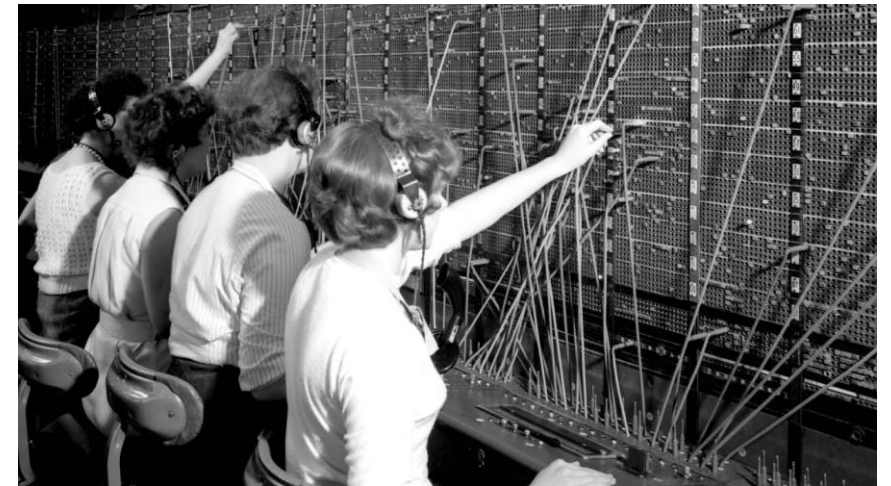
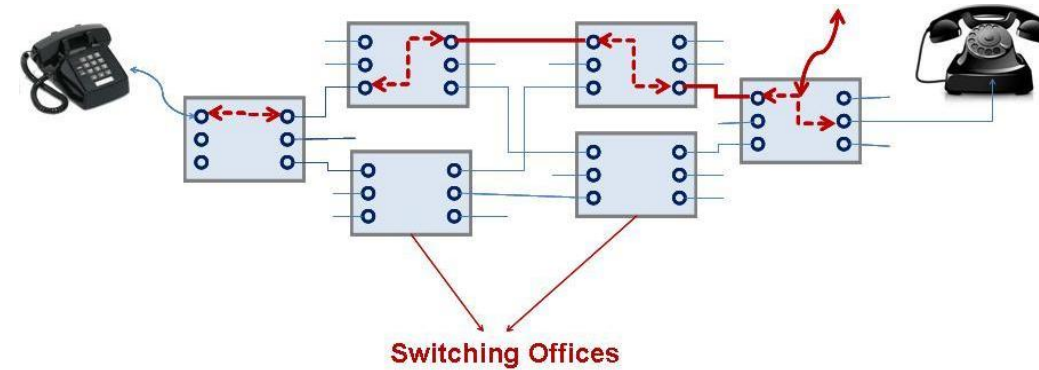
- Disparate networks onto **one platform**
- Still many points of contact and many **specialized devices** but only **one common network infrastructure**.
- A type of network that can carry **voice, video & data over the same network**



Circuit-switched network

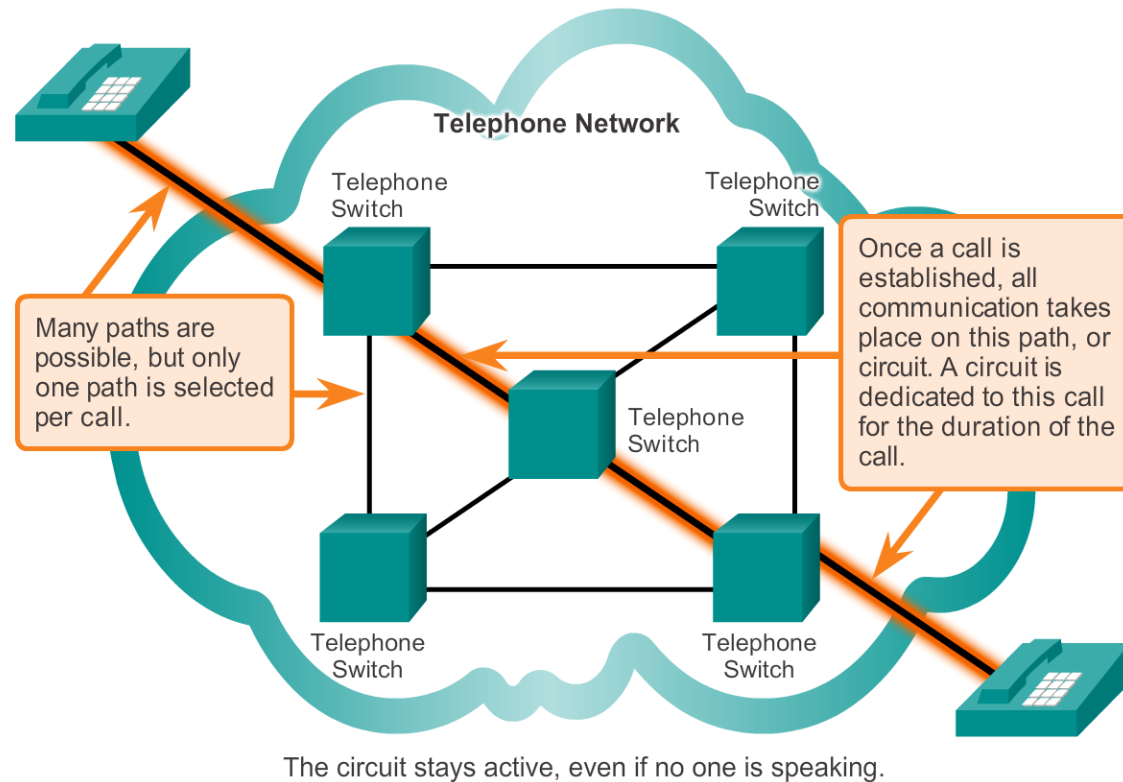
Temporary direct physical connection between two devices

- Fixed and guaranteed bandwidth
- Delivery in order
- Lost bandwidth when there is no data
- Fail in circuit → connection lost
- The circuit can not be shared with more devices
- Finite capacity to create new circuits



Circuit-switched network

Circuit Switching in a Telephone Network



There are many, many circuits, but a finite number. During peak periods, some calls may be denied.

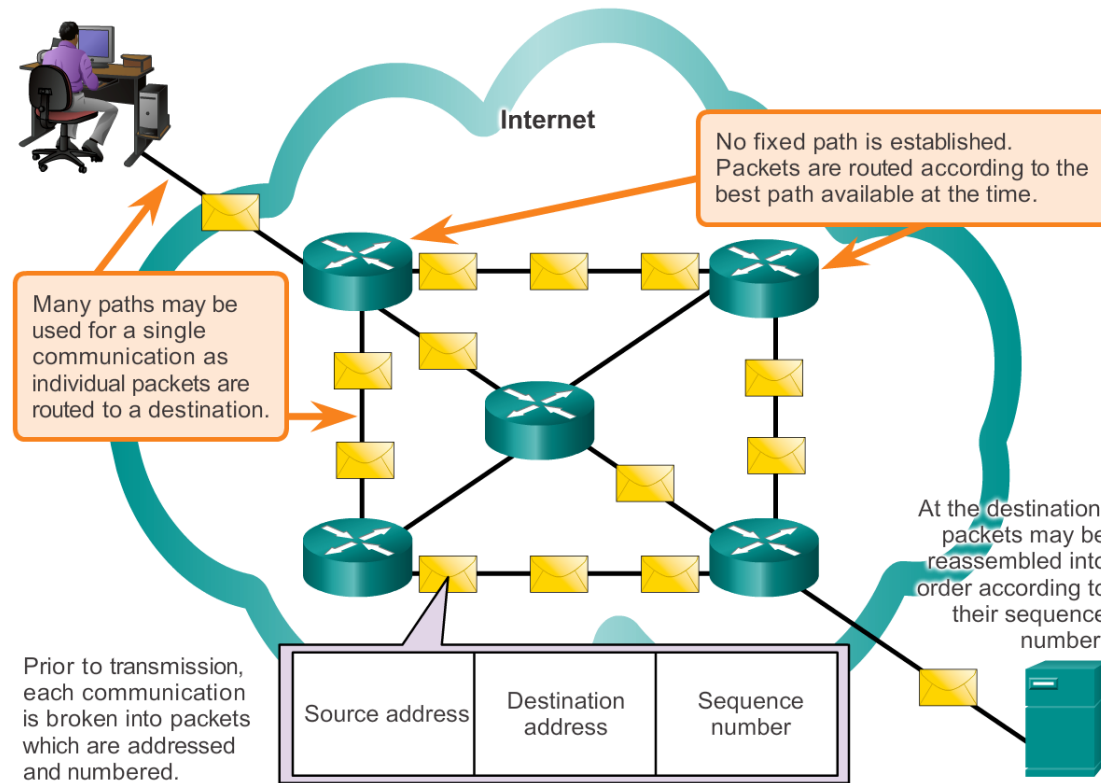
Packet-Switched Networks

Single message can be broken into multiple message blocks (=packets)
Individual blocks containing addressing information

- Can be sent through the network along various paths
- Few packets that may be lost in the advent of a failure can be retransmitted
- Network link fails → rerouting (fault tolerance)
- Medium can be shared
- No data → No bandwidth
- - Packets can arrive out of order/on different times

Packet-Switched Networks

Packet Switching in a Data Network



During peak periods, communication may be delayed, but not denied.

Reliable Network

There are four **basic characteristics** that the underlying architectures need to meet user expectations:

Fault Tolerance

- Limits the impact of a hardware or software failure
- Recovers quickly when a failure occurs
- Depend on redundant links

Scalability

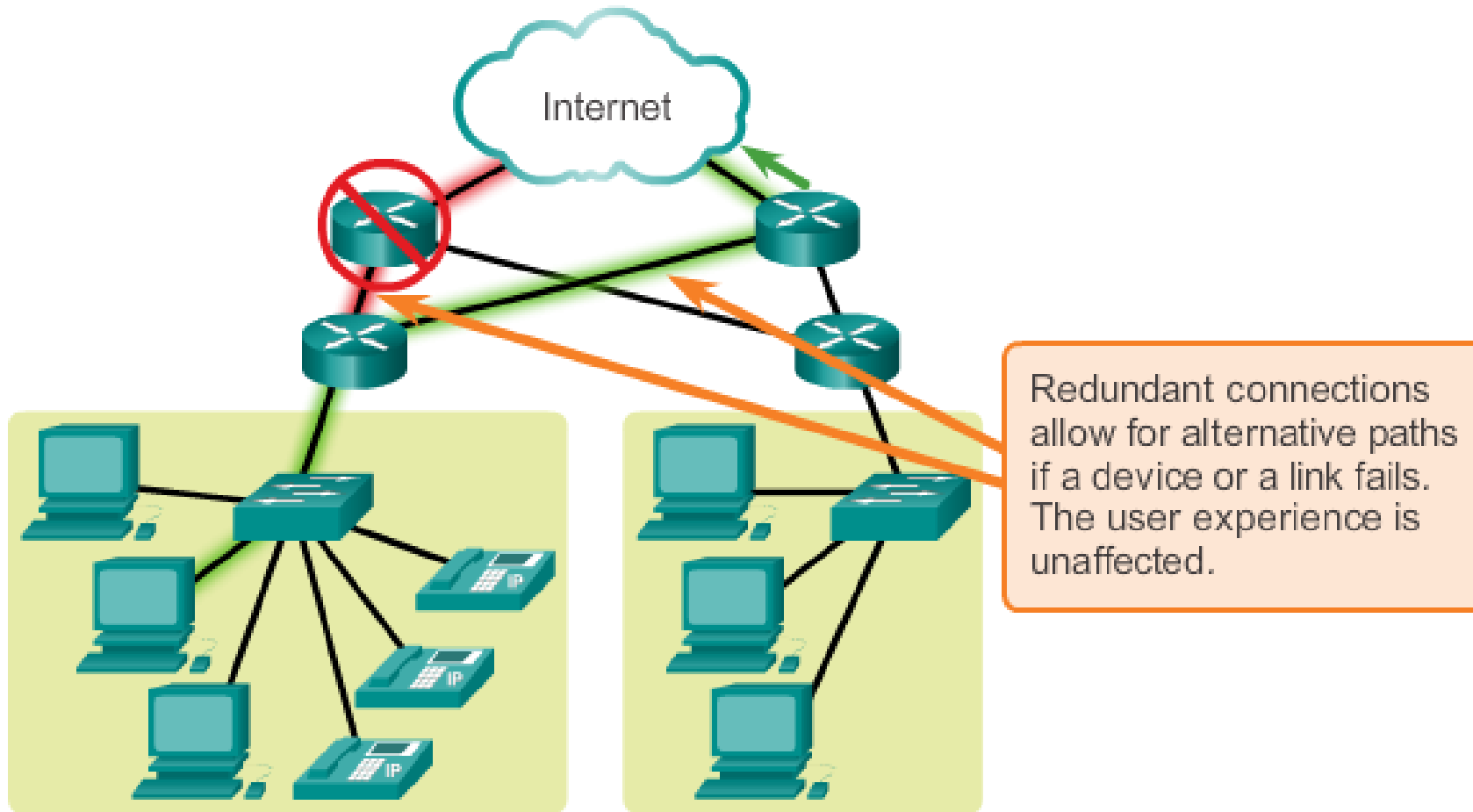
- - Expand quickly to support new users and applications

Quality of Service (QoS)

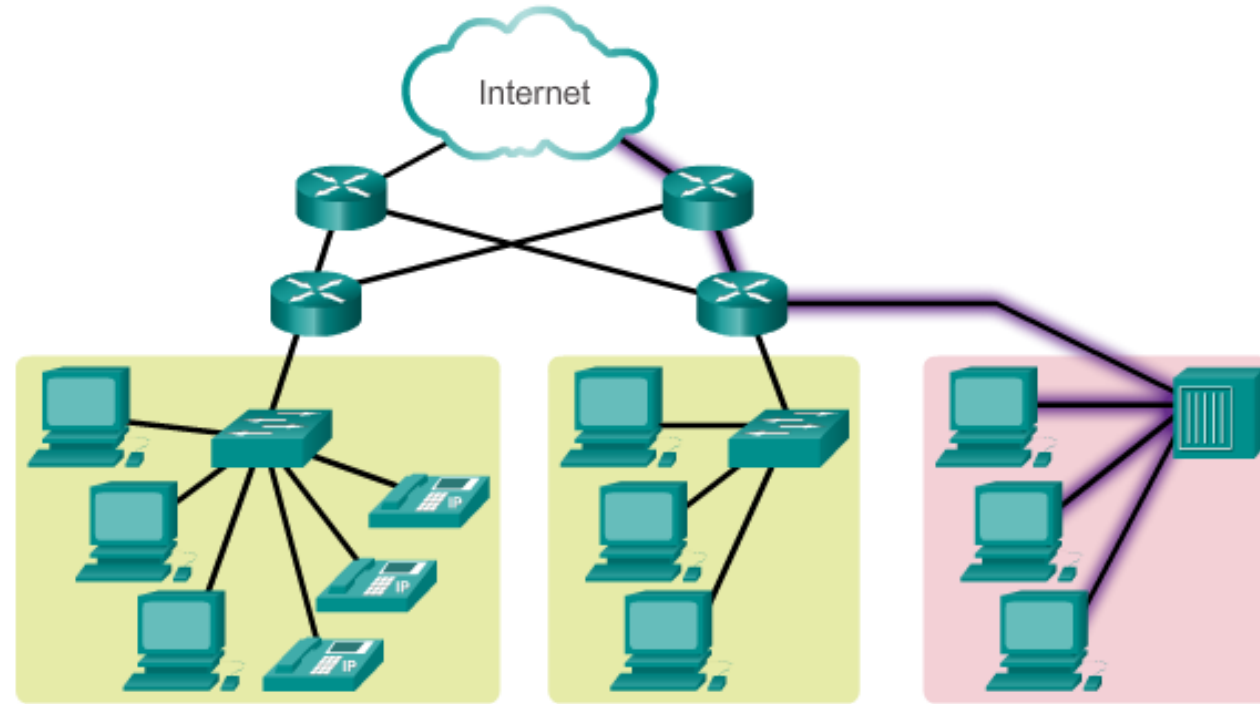
- - Voice and live video transmissions require a level of consistent quality and uninterrupted delivery

Security

Fault Tolerance



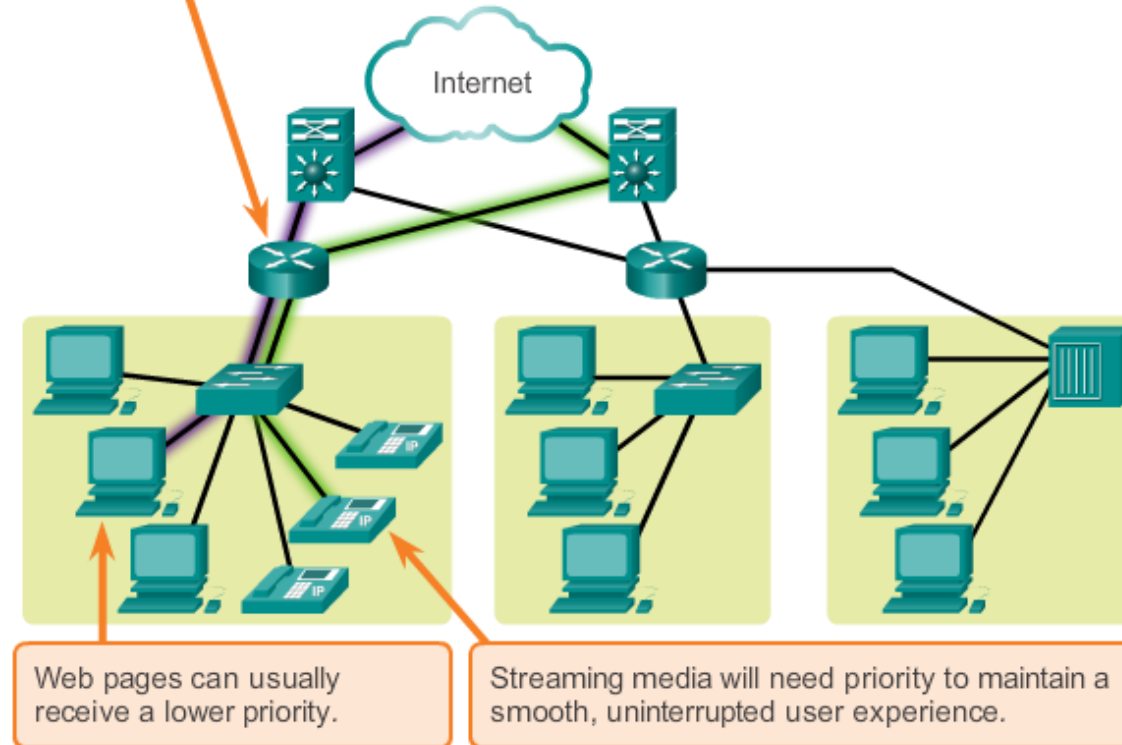
Scalability



Additional users and whole networks can be connected to the Internet without degrading performance for existing users.

Quality of Service (QoS)

Quality of Service, managed by the router, ensures that priorities are matched with the type of communication and its importance to the organization.



Security

Administrators can protect the network with software and hardware security and by preventing physical access to network devices.

